

OpenMFD turns one parameterized design into masks, printed parts, and a packaged device

A

Parameterized OpenMFD preset

OpenMFD preset

editable defaults

unit pitch	18 x 9 mm
array	6 x 8 units; 96 wells
well radius	2.50 mm
microchannels	62 at 10 um
pin / hole	1.85 / 2.00 mm
outer taper	16 deg, 3.8 mm tall
PDMS scale	x1.0226 at 100 deg C

Demonstrated-device defaults;
editable, not fixed constants.

matched
photomask DXFs

printed
insert STLs

printed
frame STL

wafer-scale
mask CAD

same coordinates
for every output

C

Plate-format wafer-mask DXF

actual v27 aligned.dxf

B

Actual single-unit photomask DXFs

single 18 x 9 mm unit

top wells

62 bottom channels

actual v27 single_top/bottom DXFs

single_bottom.dxf

single_top.dxf

top + bottom masks with wafer outline; 6 x 8 units, 96 wells

D

Generated insert STLs

single_insert.stl

wells_insert.stl

equal-aspect source projections; true z scale

pin/hole rule: 1.85/2.00 mm = 150 um total clearance (75 um/side)

E

Generated plate-frame STL

frame STL

printed frame STL

package CAD generated alongside mold outputs

F

Assembled plate-format device

framed PDMS device in microplate-compatible package